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Corrigendum

Corrigendum to "Solving $X^{2^{3n}+2^{2n}+2^n-1}+(X+1)^{2^{3n}+2^{2n}+2^n-1}=b$ in $\mathbb{F}_{2^{4n}}$ and an alternative proof of a conjecture on the differential spectrum of the related monomial functions" [Finite Fields Appl. 83 (2022) 102086]

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ABSTRACT

Let $\mathfrak{S}_2$ be the set of b's such that the equation $X^{2^{3n}+2^{2n}+2^n-1}+(X+1)^{2^{3n}+2^{2n}+2^n-1}=b$ has two solutions in $\mathbb{F}_{2^{4n}}$. This set was originally defined in the article ["Solving $X^{2^{3n}+2^{2n}+2^n-1}+(X+1)^{2^{3n}+2^{2n}+2^n-1}=b$ in $\mathbb{F}''_{2^{4n}}$ ([1]) and an alternative proof of a conjecture on the differential spectrum of the related monomial functions. *Finite Fields and Their Applications*, 83, 102086, 2022]. We show that the definition of $\mathfrak{S}_2$ (initially considered from our computational experiments for $n \leq 8$) in [1] was, unfortunately, introduced incorrectly. We, therefore, provide in this note the necessary corresponding corrections.

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Polynomial
APN function
Differential Uniformity
Symmetric cryptography

More precisely, we present the correct versions of Theorem 2 and Corollary 11, respectively.
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Let k and l be two integers such that l is a divisor of k. Define the norm $\mathbf{N}_l^k$ and trace $\mathbf{Tr}_l^k$ mappings from the finite field $\mathbb{F}_{2^k}$ to the finite field $\mathbb{F}_{2^l}$ by

$$\mathbf{N}_l^k(x) := \prod_{i=0}^{\frac{k}{l}-1} x^{2^{li}}, \quad \mathbf{Tr}_l^k(x) := \sum_{i=0}^{\frac{k}{l}-1} x^{2^{li}},$$

respectively.

Let n be a positive integer and $q = 2^n$. Theorem 2 in [1], which is the main result of [1], states that the equation $X^{2^{3n}+2^{2n}+2^n-1} + (X+1)^{2^{3n}+2^{2n}+2^n-1} = b$ has two solutions in $\mathbb{F}_{2^{4n}}$ if and only if b is in

$$\left\{ b \in \mathbb{F}_{q^4} \setminus \mathbb{F}_{q^2} \mid \mathbf{Tr}_1^n \left(\mathbf{Nr}_n^{4n} \left(\frac{b^{q+1}+1}{b+b^{q^2}} \right) \right) = 1 \right\}. \tag{1}$$

From Lemma 5, Lemma 7 and the proof of Lemma 9 in [1], we know that the cardinality of that set should be $\frac{q^4-q^3}{2}$, provided that the definition of the set is correct.

We shall employ the following propositions given in [1] to prove the unfortunate flaw of that definition (initially found from our computational experiments for $n \leq 8$).

Proposition 1 *([1, Proposition 3]). Let m be a positive integer. Then, every element z of $\mathbb{F}_{2^m}^* := \mathbb{F}_{2^m} \setminus \{0\}$ can be written twice as $z = c + \frac{1}{c}$ where $c \in \mathbb{F}_{2^m}^\star := \mathbb{F}_{2^m} \setminus \mathbb{F}_2$ if $\mathbf{Tr}_1^m(\frac{1}{z}) = 0$ and $c \in \mu_{2^m+1}^\star := \{\zeta \in \mathbb{F}_{2^m} \mid \zeta^{2^m+1} = 1\} \setminus \{1\}$ if $\mathbf{Tr}_1^m(\frac{1}{z}) = 1$.* □

Proposition 2 *([1, Lemma 4]). For a positive integer $m \geq 1$, define $\mu_m := \{x \in \mathbb{F}_{q^4} \mid x^m = 1\}$. Then, $\mathbb{F}_{q^4} \setminus \{0\} = \mu_{q-1} \cdot \mu_{q+1} \cdot \mu_{q^2+1}$.* □

The following statement shows that the set (1), in fact, coincides with $\mathbb{F}_{q^4} \setminus \mathbb{F}_{q^2}$.

Proposition 3. *For every $b \in \mathbb{F}_{q^4} \setminus \mathbb{F}_{q^2}$, it holds*

$$\mathbf{Tr}_1^n \left(\mathbf{Nr}_n^{4n} \left(\frac{b^{q+1}+1}{b+b^{q^2}} \right) \right) = 1.$$

Proof. We can set $b = z \cdot \lambda \cdot t$ where $(z, \lambda, t) \in \mu_{q-1} \times \mu_{q+1} \times \mu_{q^2+1}$. Let $M = z + z^{-1}$, $T_i = t^i + t^{-i}$ for $i \geq 0$ and $T = T_1$. It is easy to check $T_{q-1}^q = T_{q+1}$, $T_{q+1} + T_{q-1} = T^{q+1} \in \mathbb{F}_q$ and $T_{q+1} \cdot T_{q-1} = T^{2q} + T^2$. Then,

$$\begin{aligned}
&\mathbf{Tr}_1^n\left(\mathbf{Nr}_n^{4n}\left(\frac{b^{q+1}+1}{b+b^{q^2}}\right)\right)=\mathbf{Tr}_1^n\left(\mathbf{Nr}_n^{4n}\left(\frac{z^2t^{q+1}+1}{\lambda z(t+t^{-1})}\right)\right)\\
&=\mathbf{Tr}_1^n\left(\frac{(z^2t^{q+1}+1)(z^2t^{-q-1}+1)(z^2t^{q-1}+1)(z^2t^{1-q}+1)}{z^4T^{2(q+1)}}\right)\\
&=\mathbf{Tr}_1^n\left(\frac{(M^2+T_{q+1})(M^2+T_{q-1})}{T^{2(q+1)}}\right)=\mathbf{Tr}_1^n\left(\frac{M^2}{T^{q+1}}+\left(\frac{M^2}{T^{q+1}}\right)^2+\frac{1}{T^2}+\frac{1}{T^{2q}}\right)\\
&=\mathbf{Tr}_1^n\left(\frac{1}{T^2}+\frac{1}{T^{2q}}\right)=\mathbf{Tr}_1^{2n}\left(\frac{1}{T^2}\right)\overset{Prop.\ 1}{=}1. \quad \square
\end{aligned}$$

The mistakes in the main results of [1] were caused due to the incorrect statement "One can also confirm that Equation (25) is satisfied if Equation (35) and Equation (36) are true by plotting the above discussion back" (localized at Page 16 in [1], the last part of the proof of Lemma 9). As easily recognized, first of all, this statement should be repaired as "One can also confirm that Equation (25) is satisfied if Equation (34) and Equation(35) are true by plotting the above discussion back." Note that (36) in [1] defines just (1), while Equation (34) and Equation (35) are given respectively in [1] as

$$t^{2q}=\frac{Ut^2+U+1}{(U+1)t^2+U} \tag{2}$$

and

$$T=\frac{1+\delta^q}{\sqrt{U+U^2}}, \tag{3}$$

respectively, where $U:=\frac{(c^{q^2+q}+1)^{q^2+1}}{(c+c^{q^2})^{q+1}}$ and $\delta:=\left(\frac{c+c^{q^2}}{c^{q^2+1}}\right)^{q-1}$ with $c:=\frac{1}{\sqrt{b}}$.

Then, $\mathfrak{S}_2$ should be defined as the set of b's such that Equation (2) and Equation (3) hold simultaneously.

We derive: (2) $\iff t^{2q+2}(U+1)+t^{2q}U+t^2U+(U+1)=0 \iff \left(\frac{t^{q+1}+1}{t+t^q}\right)^2=\frac{U}{U+1} \iff \frac{1}{t+t^q}+\left(\frac{1}{t+t^q}\right)^{q^2}=\sqrt{\frac{U}{U+1}} \iff \frac{T+T^q}{(t+t^q)^{q^2+1}}=\sqrt{\frac{U}{U+1}} \iff \frac{T+T^q}{(t+t^q)(t^{-1}+t^{-q})}=\sqrt{\frac{U}{U+1}} \iff \frac{T+T^q}{T_{q-1}}=\sqrt{\frac{U}{U+1}}$. Since $T+T^q=\frac{\delta+\delta^{-1}}{\sqrt{U(U+1)}}$, that condition is equivalent to

$$T_{q-1}=\frac{\delta+\delta^{-1}}{U}. \tag{4}$$

Now, we need an expression of T_{q-1} in terms of T.

Proposition 4. *Let $\{T_i\}_{i\geq 0}$ be a sequence in a binary field, defined by the linear recurrence relation $T_{i+1}=T\cdot T_i+T_{i-1}$ and the initial conditions $T_0=0, T_1=T$. Then, for every $k\geq 0$, it holds $T_{2k}={T_k}^2$ and $T_{2^k-1}=T\cdot\left(1+\sum_{i=1}^{k-1}T^{\sum_{j=i}^{k-1}2^j}\right)$.*

Proof. Clearly, we have $T_2 = T^2$ and $T_4 = T^4$. Assume that $T_{2j} = {T_j}^2$ holds for every $j \leq k$. Then by the definition of linear recurrence, $T_{2(k+1)} = T \cdot T_{2k+1} + T_{2k} = T \cdot (T \cdot T_{2k} + T_{2k-1}) + T_{2k} = T \cdot (T \cdot T_{2k} + \frac{T_{2k}+T_{2(k-1)}}{T}) + T_{2k} = T^2 \cdot T_{2k} + T_{2(k-1)} = (T \cdot T_k + T_{k-1})^2 = T_{k+1}^2$. Thus, for every $k \geq 0$, $T_{2k} = {T_k}^2$.

Therefore when $k = 2$, $T_3 = T + T^3$, i.e., the equality is correct. Assuming $T_{2^k-1} = T \cdot \left(1 + \sum_{i=1}^{k-1} T^{\sum_{j=i}^{k-1} 2^j}\right)$, we have $T_{2^{k+1}-1} = \frac{T_{2(2^k-1)}+T_{2^k+1}}{T} = \frac{T_{2^k-1}^2+T^{2^{k+1}}}{T} = T \cdot \left(1 + \sum_{i=1}^{k-1} T^{\sum_{j=i+1}^{k} 2^j}\right) + T^{2^k+2^{k-1}+\cdots+1} = T \cdot \left(1 + \sum_{i=1}^{k} T^{\sum_{j=i}^{k} 2^j}\right)$. $\square$

Since $T = \frac{1+\delta^q}{\sqrt{U+U^2}}$, one has $T^q = \delta T$ and therefore Condition (4), or equivalently, $T \cdot \left(1 + \sum_{i=1}^{k-1} T^{\sum_{j=i}^{k-1} 2^j}\right) = \frac{\delta+\delta^{-1}}{U}$, can be rewritten as

$$T^q + T^{q+q/2} + \cdots + T^{q+q/2+\cdots+2} = \frac{\delta^2+1}{U}.$$

Taking the square root to both sides of this equality gives $T^{q/2} + T^{q/2+q/4} + \cdots + T^{q/2+q/4+\cdots+1} = \frac{\delta+1}{\sqrt{U}}$, or equivalently, $T^q \cdot \mathrm{Tr}_1^n(\frac{1}{T}) = \frac{\delta+1}{\sqrt{U}}$, i.e.,

$$\mathbf{Tr}_1^n\left(\frac{1}{T}\right) = \sqrt{U+1}.$$

By substituting $\frac{1}{T} = \frac{c^{q^2+1}}{c+c^{q^2}} \cdot \frac{\sqrt{\mathbf{N}_n^{4n}(c^{q+1}+1)}}{\mathbf{Tr}_n^{4n}\left(c^{q^2+q+1}\right)}$ and $U + 1 = \frac{(c^{q+1}+1)^{q^2+1}}{(c+c^{q^2})^{q+1}}$, one has $\mathbf{Tr}_1^n\left(\frac{c^{q^2+1}}{c+c^{q^2}} \cdot \frac{\sqrt{\mathbf{N}_n^{4n}(c^{q+1}+1)}}{\mathbf{Tr}_n^{4n}\left(c^{q^2+q+1}\right)}\right) = \sqrt{\frac{(c^{q+1}+1)^{q^2+1}}{(c+c^{q^2})^{q+1}}}$. By again substituting $c = \frac{1}{\sqrt{b}}$, one gets the following corrected versions of Theorem 2 and Corollary 11 in [1], respectively.

Theorem 5. *Define*

$$\mathfrak{S}_2 := \left\{ b \in \mathbb{F}_{q^4} \setminus \mathbb{F}_{q^2} \mid \mathbf{Tr}_1^n\left(\frac{\mathbf{N}_n^{4n}(b^{q+1}+1)}{\left[(b+b^{q^2})\mathbf{Tr}_n^{4n}(b)\right]^2}\right) = \frac{(b^{q+1}+1)^{q^2+1}}{(b+b^{q^2})^{q+1}} \right\},$$

with the convention $\frac{\cdot}{0} := 0$. *Then, the number of solutions of the equation* $X^{2^{3n}+2^{2n}+2^n-1} + (X+1)^{2^{3n}+2^{2n}+2^n-1} = b$ *in* $\mathbb{F}_{q^4}$ *is equal to*

1. q^2 *if* $b = 1$,
2. $q^2 - q$ *if* $b \in \mu_{q+1}^\star$,
3. 2 *if* $b \in \mathfrak{S}_2$.
4. 0 *if* $b \in \mathbb{F}_{q^4} \setminus \{\mathfrak{S}_2 \cup \mu_{q+1}^\star \cup \{1\}\}$. $\square$

Corollary 6. *With convention* $\frac{\cdot}{0} := 0$*, it holds*

$$\#\left\{b \in \mathbb{F}_{q^4} \setminus \mathbb{F}_{q^2} \mid \mathbf{Tr}_1^n\left(\frac{\mathbf{N}_n^{4n}(b^{q+1}+1)}{\left[(b+b^{q^2})\mathbf{Tr}_n^{4n}(b)\right]^2}\right) = \frac{(b^{q+1}+1)^{q^2+1}}{(b+b^{q^2})^{q+1}}\right\} = \frac{q^4-q^3}{2}. \quad \square$$

Data availability

No data was used for the research described in the article.

References

[1] K.H. Kim, S. Mesnager, Solving $X^{2^{3n}+2^{2n}+2^n-1}+(X+1)^{2^{3n}+2^{2n}+2^n-1} = b$ in $\mathbb{F}_{2^{4n}}$ and an alternative proof of a conjecture on the differential spectrum of the related monomial functions, Finite Fields Appl. 83 (2022) 102086, https://doi.org/10.1016/j.ffa.2022.102086.